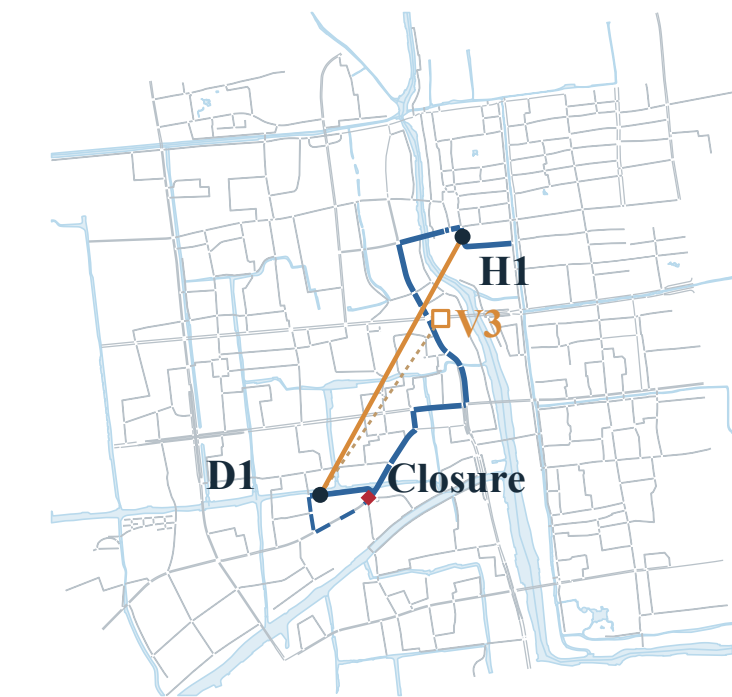


Contrasting network morphology

(a) Site A: Suzhou



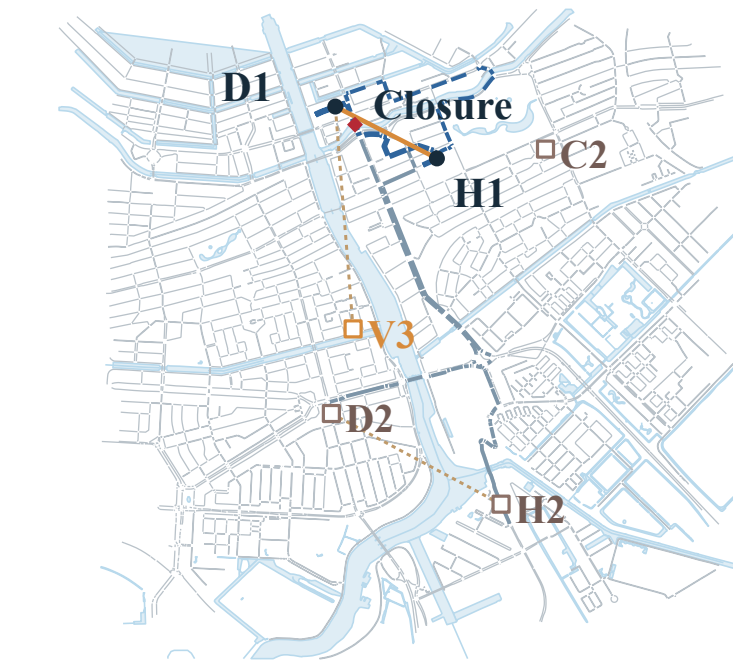
500 m

Urban (meshed)

Primary ground

Detour

(b) Site B: Amsterdam

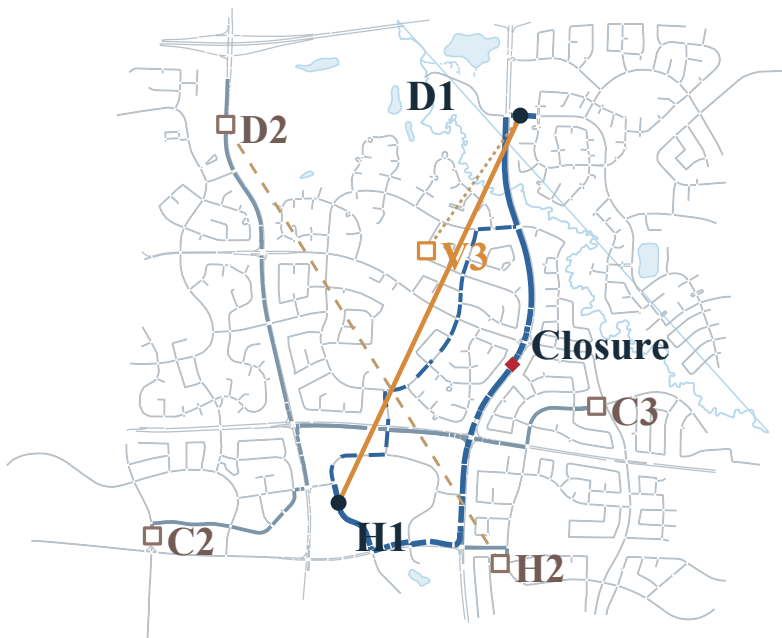


500 m

River (barrier)

Primary air

(c) Site C: Edmonton



500 m

Suburban (sparse)

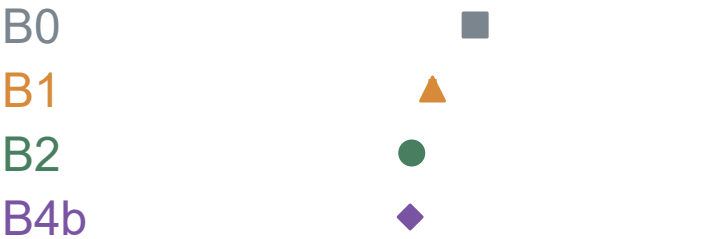
Auxiliary ground

Other air

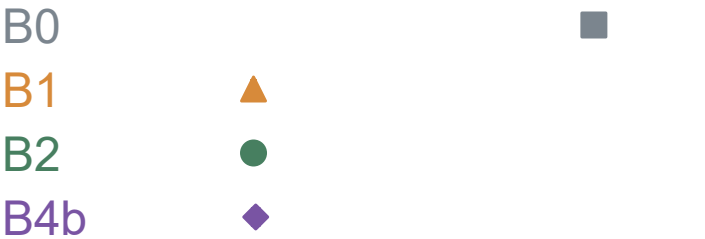
a Completion time

Mean and 95% interval

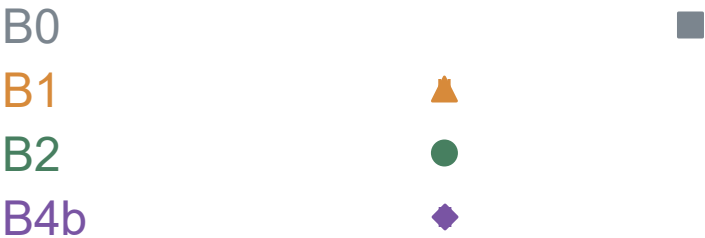
Site A



Site B



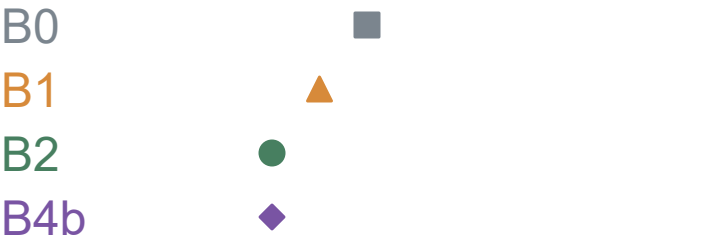
Site C



b Missed deadlines

Mean and 95% interval

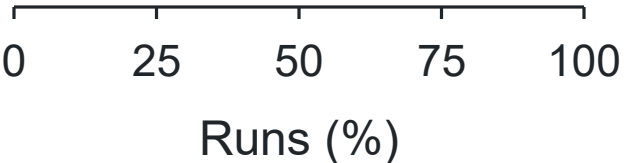
Site A



Site B



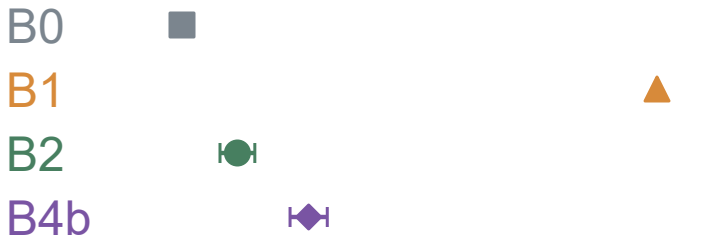
Site C



c Incumbent damage

Mean and 95% interval

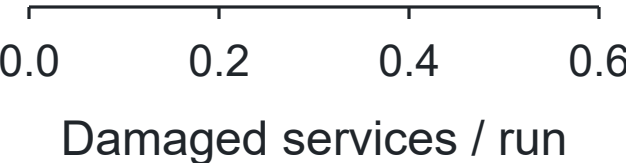
Site A

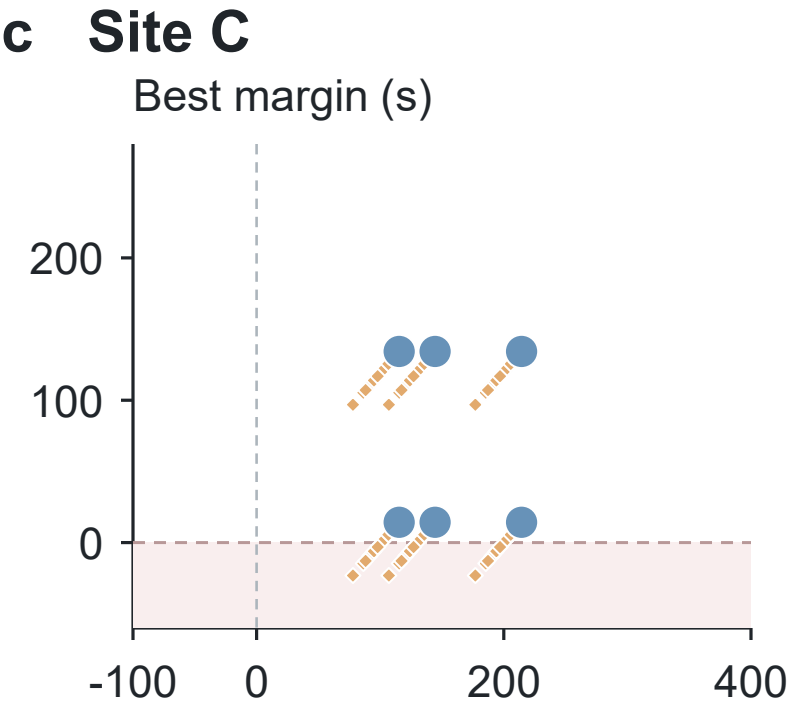
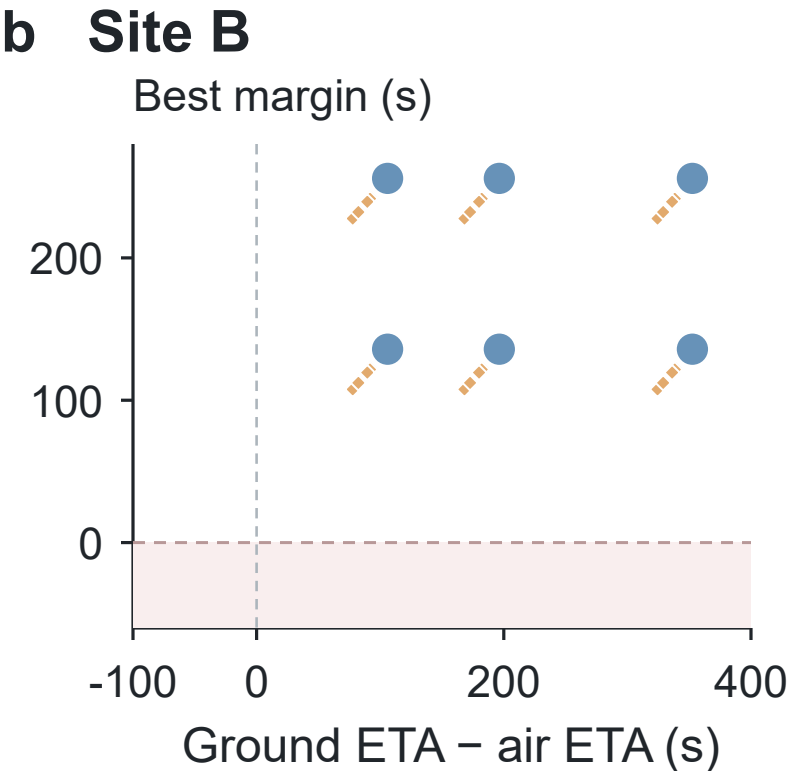
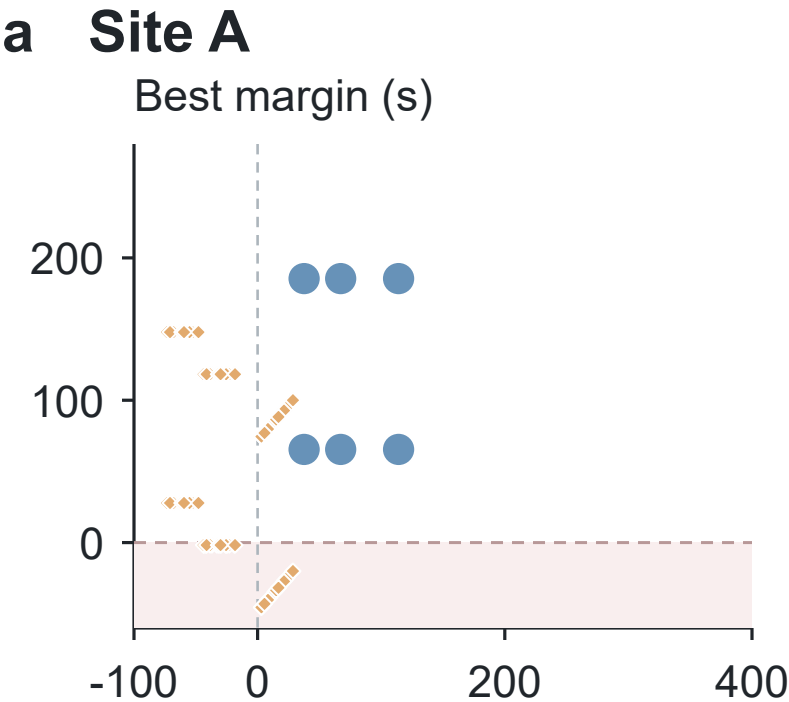


Site B



Site C





(d) Air selection (%)

	B1	B2	B4b	
A	100	67	67	n=180
B	100	0	60	n=20
C	100	35	50	n=40

(e) Air selection (%)

	B1	B2	B4b	
A	100	100	100	n=180
B	100	100	95	n=60
C	—	—	—	n=0

(f) Air selection (%)

	B1	B2	B4b	
A	100	100	100	n=193
B	100	100	100	n=20
C	100	100	100	n=27

A Other timely states B Faster preemptive air vs timely ground C No timely option

◆ Preemptive air ● Other air

a Site A



Path restored / affected

B1 200 / 200
B2 200 / 200
B4b 191 / 191

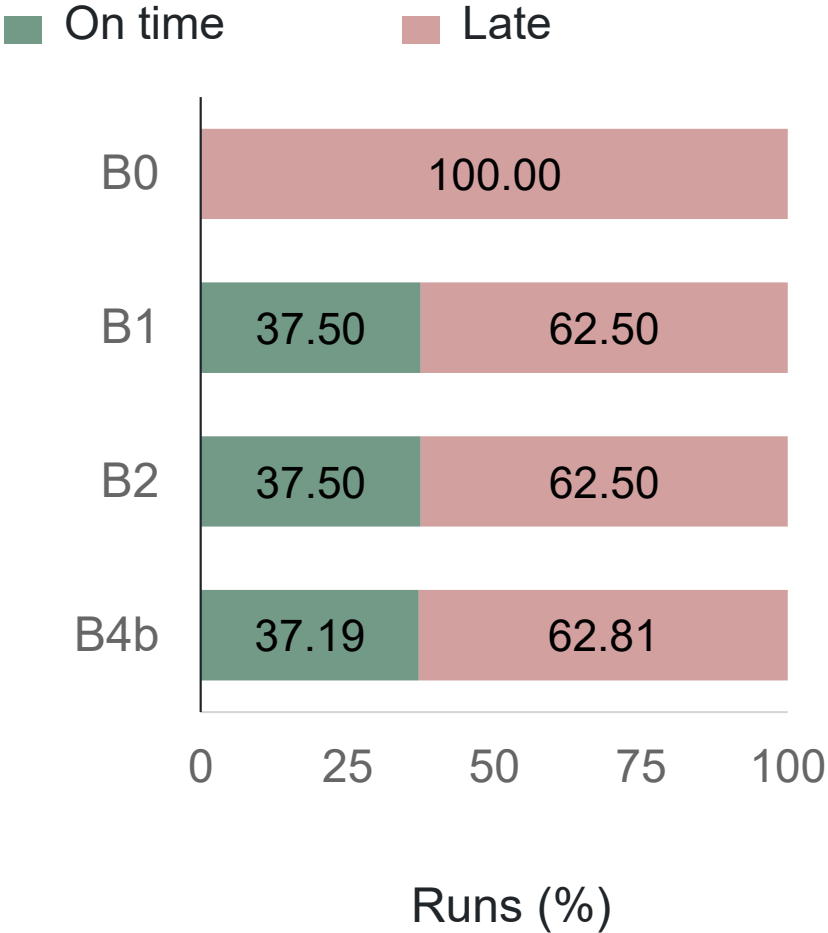
b Site B



Path restored / affected

B1 200 / 200
B2 200 / 200
B4b 199 / 199

c Site C



Path restored / affected

B1 200 / 200
B2 200 / 200
B4b 201 / 201

a Site A

	L1	L2	L3	L4
B0	240	240	240	240
B1	0	240	0	120
B2	0	240	0	120
B4b	6	240	1.7	123.4

b Site B

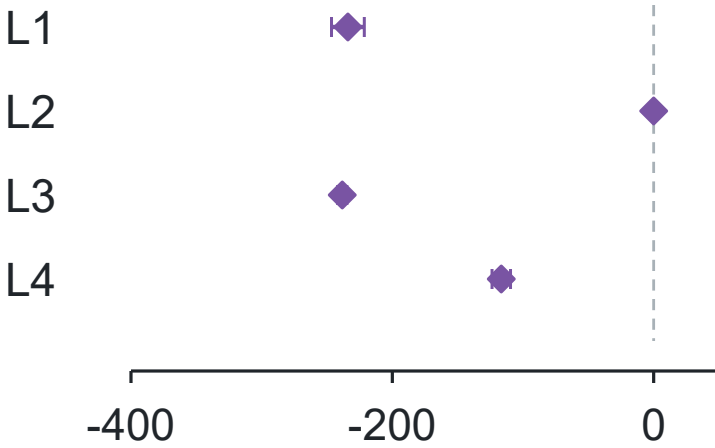
	L1	L2	L3	L4
B0	240	240	375	375
B1	0	120	135	255
B2	0	120	135	255
B4b	0	120	135	255

c Site C

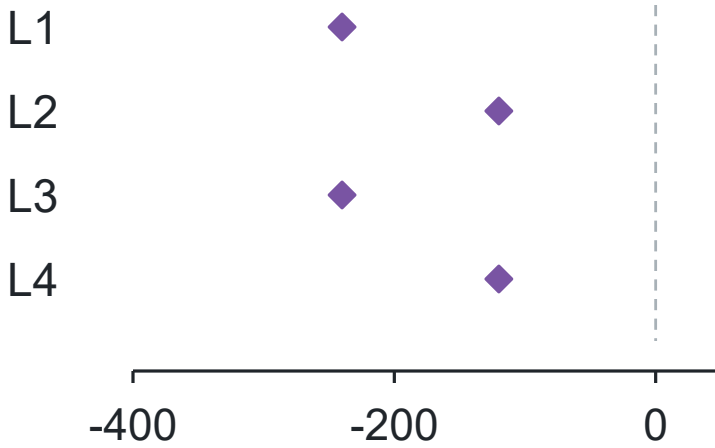
	L1	L2	L3	L4
B0	240	240	375	375
B1	240	240	135	375
B2	240	240	135	375
B4b	240	240	135	375



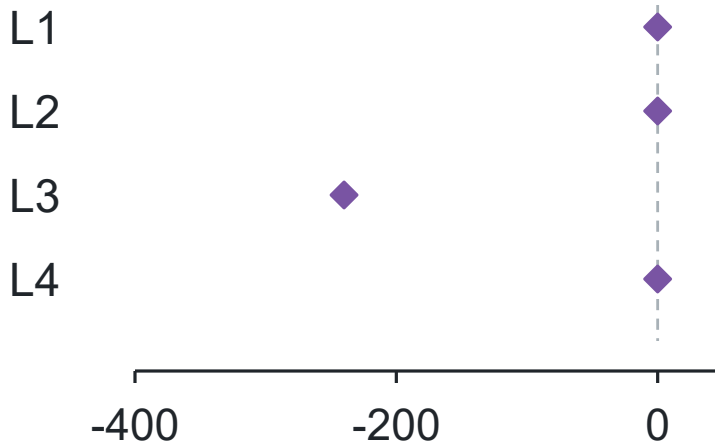
d B4b – B0



e B4b – B0

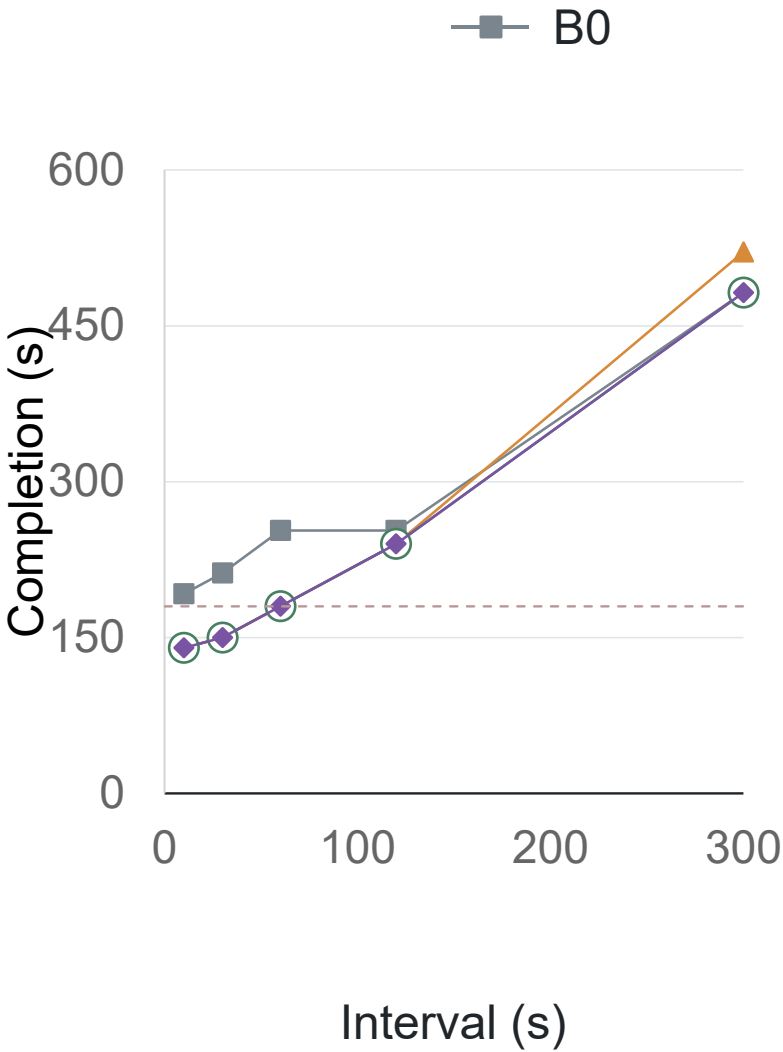


f B4b – B0



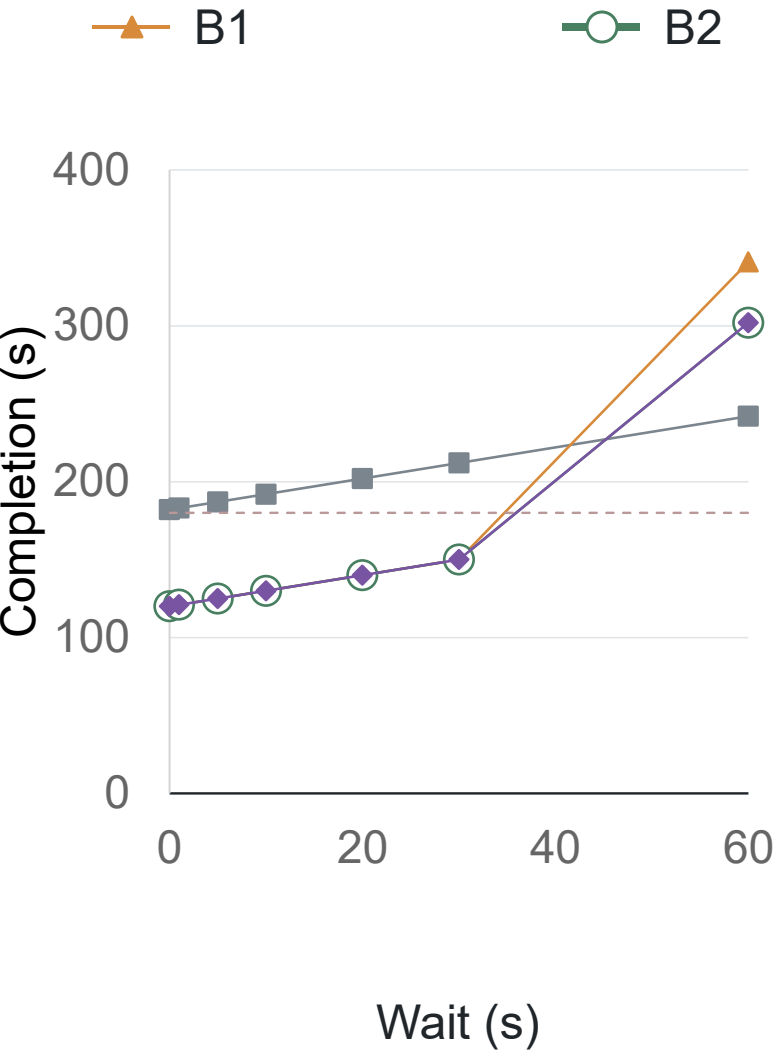
Paired change in SWL (negative values indicate lower loss)

a Observation interval



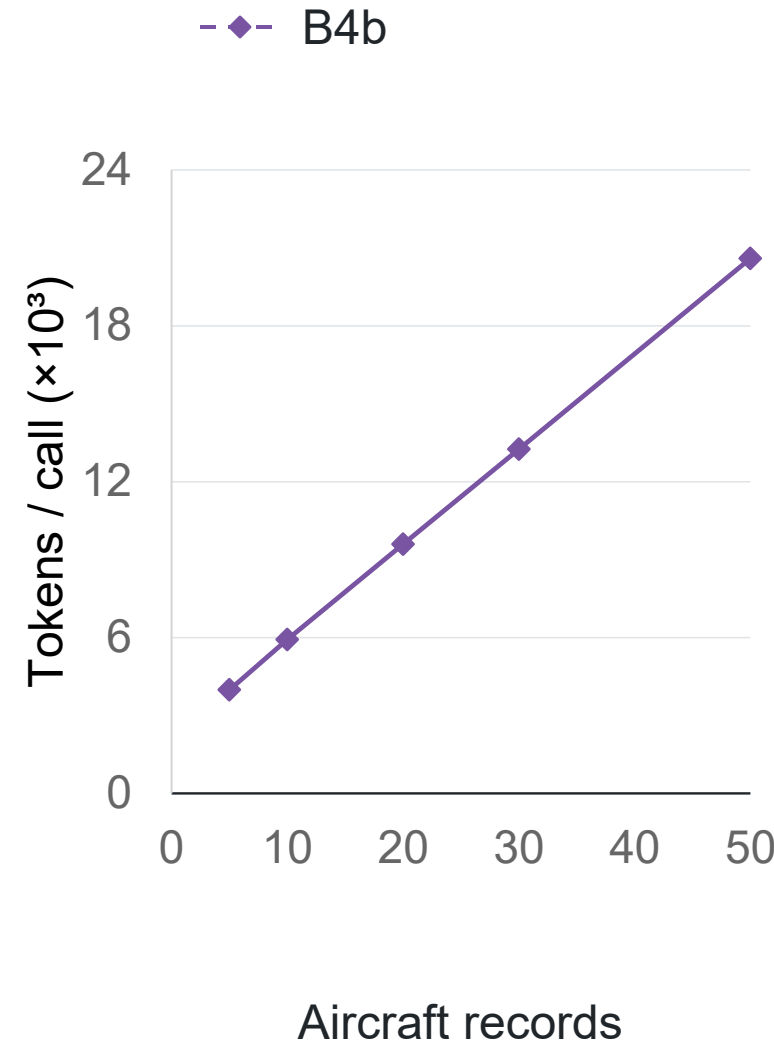
B2 = B4b at every interval
180 s service window

b Execution wait



B1 = B2 = B4b at 0–30 s
B2 = B4b at 60 s

c Input burden



Four active aircraft
5–50 total records

a B0: repeated ground command



b B4b: air replaced by ground

